FEDERAL SPECIFICATION

ANTIPYRINE AND ETHYL AMINOBENZOATE SOLUTION (EAR DROPS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers antipyrine and ethyl aminobenzoate solution suitable for instillation in the human ear in the treatment of the common earache.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specification and standards.**—The following specification and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

PPP–M–00186—Medicinal Products and Clinical Laboratory Reagents; General Specification for Containers (Packaging and Packing).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Pharmaceutical Association:

National Formulary

(Application for copies should be addressed to the J. B. Lippincott Company, Philadelphia, Pa.)

U. S. Pharmacopoeial Convention, Inc.:

Pharmacopoeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa.)

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3. REQUIREMENTS

3.1 Material.—Antipyrine and ethyl aminobenzoate solution shall be a clear, colorless to light amber liquid in accordance with the following formula:

Glycerin, U.S.P. (anhydrous)	17.90 g.
Antipyrine, N.F.	0.81 g.
Ethyl aminobenzoate, N.F.	0.21 g.

Ethyl aminobenzoate content shall be not less than 94 percent and not more than 106 percent of the specified amount when assayed as specified in 4.5.1 and 4.5.1.1. Antipyrine content shall be not less than 94 percent and not more than 106 percent of the specified amount when assayed as specified in 4.5.1 and 4.5.1.2. The residue obtained in each respective assay shall comply with the tests specified in 4.5.1.3.

3.1.1 *Glycerin.*—The glycerin shall be in accordance with the tests, standards, and requirements of the U.S.P. and shall be anhydrous. The glycerin shall be sufficiently dehydrated so that the finished product shall comply with 3.2.3.

3.1.2 *Antipyrine.*—The antipyrine shall be in accordance with the tests, standards, and requirements of the N.F.

3.1.3 *Ethyl aminobenzoate.* — The ethyl aminobenzoate shall be in accordance with the tests, standards, and requirements of the N.F.

3.2 Chemical and physical characteristics of the antipyrine and ethyl aminobenzoate solution.

3.2.1 *Identity.* — The color shall become greenish-brown when tested as specified in 4.5.2.

3.2.2 *Spectrophotometric analysis for color control.*—The absorbance (optical density) shall be not greater than 0.700 when tested as specified in 4.5.3.

3.2.3 *Specific gravity.*—The specific gravity shall be not less than 1.250 and not more than 1.255 at 25° centigrade (C.) when tested as specified in 4.5.4.

3.2.4 *pH value.*—The pH value shall be between 4.5 and 5.0 at 20° C. when tested as specified in 4.5.5.

3.2.5 *Optical rotation.*—The angle of rotation at 25° C. shall be between plus (+) 1.5° and minus (—) 1.5° when tested as specified in 4.5.6.

3.2.6 *Surface tension.*—The surface tension shall be between 55.0 and 55.5 dynes/cm. at 25° C. when tested as specified in 4.5.7.

3.2.7 *Stability.*—No precipitate shall form when tested as specified in 4.5.8.1 and no color change or precipitate shall develop when tested as specified in 4.5.8.2.

3.2.8 *Contents.* — Filled immediate containers shall contain an average volume of 15.0 cubic centimeters (cc.) and no one immediate container shall contain less than 14.75 cc. when tested as specified in 4.5.9.

3.3 Workmanship.—The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection for procurement. — Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by

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the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 3 years after delivery of the supplies to which such records relate.

4.2 Lot.—For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of antipyrine and ethyl aminobenzoate solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling.—Unless otherwise specified sampling shall be conducted in accordance with Military Standard MIL–STD–105 and appendix thereto.

For visual examination	Inspection level	AQL (defects per 100 units)	Unit of product
Major A	L6	1.0	Filled and labeled bottle
Major B	II (normal) (single)	1.0	Filled and labeled bottle
Major C	II (normal) (single)	1.0	Intermediate containers
Major D	II (normal) (single)	1.0	Exterior containers

For end item testing.—Acceptance number shall be zero (0) for all sample sizes.

Characteristic	Requirement paragraph	Test paragraph	Appendix level	Minimum sample
			L2 To obtain a composite sample to be utilized in performing the following tests.	5
Material	3.1	4.5.1		
Identity	3.2.1	4.5.2		
Spectro analysis	3.2.2	4.5.3		
Specific gravity	3.2.3	4.5.4		
pH	3.2.4	4.5.5		
Optical rotation	3.2.5	4.5.6		
Surface tension	3.2.6	4.5.7		
Stability	3.2.7	4.5.8.1	L1 *	2
		4.5.8.2	L1 *	2
Average contents	3.2.8	4.5.9	L2 *	5
Leakage	5.1.1	4.3.3 of PPP–M–00186	L3 *	10
Label adhesion	5.4.1.1.1		L3 *	10

* Where applicable, the same sample shall be used for the determination of two or more tests.

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4.4 Examination. — The antipyrine and ethyl aminobenzoate solution shall be examined to determine compliance with all requirements contained in this specification.

4.4.1 *Classification of defects.*—Examination shall be conducted in accordance with the following classification of defects:

Categories and defects**	
Major A	
101.	Antipyrine and ethyl aminobenzoate solution not clear.
102.	Antipyrine and ethyl aminobenzoate solution not uniform color specified.
103.	Antipyrine and ethyl aminobenzoate solution not free of precipitate or undissolved solids.
104.	Dropper does not withdraw and does not retain fluid.
105.	Pipette tip not smooth.
106.	Closure C not as specified.
Major B	
151.	Bottle not light resistant or not dark blue.
152.	Nipple rubber not fresh stock (indicated by cracks, splits).
153.	Labeling not complete.
Major C	
175.	Material and marking of intermediate container not as specified.
Major D	
185.	Material and marking of exterior container not as specified.

** Examination is not restricted to the classified possible defects listed above.

4.5 Tests.

4.5.1 *Assays for antipyrine and ethyl aminobenzoate.*

Reagent:

Ether mixture.—Mix equal parts of ethyl ether (which has been washed once with one-fifth of its volume of distilled water) and petroleum ether.

Accurately weigh approximately 5 g. of the sample into a separatory funnel and dilute the sample with 20 cc. of distilled water. Extract 4 times with 15 cc. of the ether mixture. The ether mixture layer contains the ethyl aminobenzoate and the aqueous solution contains the antipyrine.

4.5.1.1 *Ethyl aminobenzoate determination.* — Combine the ether extracts in a tared beaker and evaporate to dryness on a water bath. Dry in a sulfuric acid desiccator to constant weight and accurately weigh.

4.5.1.2 *Antipyrine determination.*—Transfer the aqueous solution to a beaker and heat on a water bath to remove any ether. Quantitatively transfer to a separatory funnel. Add 1 to 2 cc. of 1/10 *N* sodium hydroxide (to make alkaline) and extract 3 times with 20 cc. of chloroform. Collect the chloroform in a tared beaker and evaporate the chloroform by heating to 65° C. in a stream of air. Dry in a sulfuric acid desiccator to constant weight and accurately weigh.

4.5.1.3 *Identification and melting range.*—The residues obtained in tests 4.5.1.1 and 4.5.1.2 shall comply with the "Identification" and "Melting range" tests in the respective monographs in the N.F. for ethyl aminobenzoate and antipyrine.

4.5.2 *Identity.*—Place 1-2 drops of ferric chloride T.S. into 15 cc. of the antipyrine and ethyl aminobenzoate solution, mix, and allow to stand for 30 minutes. The coloration indicates that the antipyrine is completely integrated into the antipyrine and ethyl aminobenzoate solution.

4.5.3 *Spectrophotometric analysis for color control.*—Using a suitable spectrophotometer with a 1 centimeter (cm.) cell, the absorbance (optical density) of the antipyrine and ethyl aminobenzoate solution is recorded at 420 millimicrons, employing glycerin, U.S.P., as the blank.

4.5.4 *Specific gravity.*—The specific gravity shall be determined using a pycnometer or Specific Gravity Balance.

4.5.5 *pH value.*—Dilute the antipyrine and ethyl aminobenzoate solution with equal parts of distilled water and determine the pH potentiometrically using a glass electrode.

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4.5.6 *Optical rotation.*—The optical rotation shall be determined, in a 40 cm. tube, with a suitable polarimeter at 25° C., using the "D" line of sodium light.

4.5.7 *Surface tension.* — Surface tension shall be determined using a Du Nouy Tensiometer platinum ring.

4.5.8 *Stability.*

4.5.8.1 Store 15 cc. of the antipyrine and ethyl aminobenzoate solution at 0° C. for 48 hours.

4.5.8.2 Transfer 15 cc. of the antipyrine and ethyl aminobenzoate solution to a beaker and heat at 100° C. for one hour.

4.5.9 *Contents.*—The volume of contents shall be determined by pouring the material into a graduated cylinder calibrated to at least 0.2 cc. The volume of contents shall be determined at 25° C. ± 2° C. As an alternate, the volume of contents may be determined by weight and then converted to volume.

5. PREPARATION FOR DELIVERY

5.1 Immediate containers (see 6.1).

5.1.1 *For Military Medical Supply Agency procurement.*—Immediate containers shall be in accordance with all applicable requirements of Federal Specification PPP–M–00186. Immediate container classification shall be group A, class 1, type e, style 5, grade 1 or dark blue glass, closure C, and with an outer seal.

5.1.2 *For other agency procurement.*—Immediate containers shall be a commercial container of the type, size, and kind commonly used for the purpose or as specified in the procurement document.

5.2 Packaging.

5.2.1 *Unit of issue.*—One bottle, as specified, constitutes one unit of issue.

5.2.2 *Packaging.*—The number of units of issue specified in table I, column 2, of Federal Specification PPP–M–00186, for the applicable weight of contents, ½ oz., indicated in column 1, shall be packaged in an intermediate container constructed in accordance with the applicable paragraph referenced in column 3.

5.3 Packing.—The number of intermediate cartons containing the total number of units of issue for the applicable weight of contents, indicated in column 4, of Federal Specification PPP–M–00186, shall be overpacked in an exterior container constructed in accordance with the requirements of the applicable paragraph referenced in column 5 (level B), or column 6 (level A), for the level of protection specified in the procurement document.

5.4 Marking.

5.4.1 *Labeling* (*see 6.1*).

5.4.1.1 *For Military Medical Supply Agency procurement.*—Immediate containers shall be labeled bearing the applicable information as specified in labeling paragraph of Federal Specification PPP–M–00186. The unit of issue, quantity, and contract number are not required. Label shall bear the date of manufacture, the caution statement as required by the Durham-Humphrey law, the statements: "KEEP WELL CLOSED IN DRY PLACE" and "DO NOT RINSE DROPPER", and the formula stated as follows:

"Each cc. contains:

Antipyrine, N.F.	54 mg.
Ethyl Aminobenzoate, N.F.	14 mg.
Glycerin (anhydrous)	1.19 g."

5.4.1.1.1 *Adhesion of label.*—Label on immediate containers shall comply with adhesion of label test as specified in Federal Specification PPP–M–00186.

5.4.1.2 *For other agency procurement.*—Immediate containers shall be commercially

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labeled or as specified in the procurement document.

5.4.2 *Intermediate containers.*—Intermediate containers shall be marked in accordance with section 5 of Military Standard MIL–STD–129. Lot (control) number shall be shown. Date of manufacture shall be shown in lieu of date packed.

5.4.3 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except that 6.2(c), 6.2(d), 6.5, and 7.5.1 shall not be applicable. Lot (control) numbers shall be shown. Date of manufacture shall be shown in lieu of date packed. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Number and date of this specification.

(*b*) Immediate container required (see 5.1).

(*c*) Labeling required (see 5.4.1).

6.2 This specification covers the following item appearing in the Federal Supply Catalog, Department of Defense Section, for Medical Materiel:

6505–598–5830 ANTIPYRINE AND ETHYL AMINOBENZOATE SOLUTION, 15 cc.

6.3 Transportation description.—Transportation description applicable to this item is:

Drugs and Medicines NOIBN

Carload minimum weight 30,000 pounds per Uniform Freight Classification.

Truckload minimum weight 30,000 pounds per National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
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Copies of this specification may be purchased for 5 cents each.

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